

## Lecture 4. Second-Order Differential Equations

**Lecture Objective:** To introduce the concept of second-order differential equations, consider their general form and initial conditions, and demonstrate their connection with real physical processes.

### Concept of a Second-Order Differential Equation

A **second-order differential equation** is an equation that involves the second derivative of an unknown function.

The general form is:

$$F(x, y, y', y'') = 0$$

If the equation is solved with respect to the second derivative, it takes the form:

$$y'' = f(x, y, y')$$

The **order of a differential equation** is determined by the highest derivative appearing in the equation.

### Solution of a Second-Order Differential Equation

A **solution** of a second-order differential equation is a function  $y=y(x)$  that has continuous derivatives up to the second order and satisfies the equation identically.

#### General Solution

The general solution of a second-order differential equation contains **two arbitrary constants**:

$$y = y(x, C_1, C_2)$$

This is because solving such an equation involves **two successive integrations**.

#### Initial Conditions. Cauchy Problem

To determine a unique solution, **two initial conditions** must be specified, for example:

$$y(x_0) = y_0, \quad y'(x_0) = y_1$$

The problem of finding a solution that satisfies these conditions is called the **Cauchy problem**.

Initial conditions have a clear physical meaning:

- $y(x_0)$  represents the initial position;
- $y'(x_0)$  represents the initial velocity.

## Physical Interpretation

Second-order differential equations naturally arise in mechanics and physics because:

- the first derivative represents velocity;
- the second derivative represents acceleration.

## Example

According to Newton's second law:  $my'' = F(x, y, y')$

Thus, the law of motion of a body is described by a **second-order differential equation**.

## The Simplest Second-Order Differential Equation

Consider the equation:  $y'' = 0$

Integrating, we obtain:  $y' = C_1$ ,  $y = C_1x + C_2$

This equation describes **uniform rectilinear motion**.

## Linear Second-Order Differential Equations

The most important class is **linear second-order differential equations**, which have the form:

$$y'' + p(x)y' + q(x)y = f(x),$$

where the functions  $p(x)$ ,  $q(x)$  and  $f(x)$  are continuous on a given interval.

If  $f(x) = 0$ , the equation is called **homogeneous**.

If  $f(x) \neq 0$ , the equation is called **non-homogeneous**.

## Importance of Second-Order Differential Equations

Second-order differential equations:

- describe oscillatory processes;
- model the motion of bodies;
- are widely used in mechanics, physics, engineering, biology, and economics;
- form the basis for studying more complex models and systems.

## Self-Assessment Questions

1. What is a second-order differential equation?
2. How many arbitrary constants does the general solution contain?
3. What initial conditions are required for a second-order equation?
4. What is the physical meaning of the second derivative?
5. What is the difference between homogeneous and non-homogeneous equations?